Four-frame global referent verification

2026-10-14 | TA-STVG + frozen STVG-R1-7B | historically exposed HC2 4 / VidSTG 4. Finite pilot only; no CURRENT promotion or fresh-confirmation claim.

Candidate and teacher execution

Case	Retrieved	Legal	Rejected	Base selected	ID swap same
HC01	7	2	5	Source	False
HC02	4	1	3	Source	True
HC03	5	3	2	Source	True
HC04	7	6	1	Source	True
V01	5	1	4	Source	True
V02	4	1	3	Source	True
V03	5	3	2	Source	True
V04	6	1	5	Source	True

Source + marked Top3 + masked Top3 uses previously sealed scores, never GT shortlist. Translate each source box by the anchor center displacement while preserving its own width/height. 25 whole hypotheses rejected for image-bound violations; no clipping. Four cases have only Source, so only four cases can test ranking alternatives.

42 frozen native-video calls: complete JSON 30/42; native ID diagnostic parse 42/42; swap consistency 7/8; changed-query choice change 8/8. Base choices all Source; invalid strict formats fall back to Source. ID-only parsing is a separate diagnostic and never changes method references.

Every candidate has four complete-scene frames, chronological ordering and actual timestamp labels. Encoder uses mean-gap uniform pacing for nonuniform source gaps; sparse constant displacement is a motion hypothesis, not established tracking. Candidate-video order always places Source first; order preference remains unisolated.

Actual final full-Tube readouts

Spatial/temporal arm	HC vIoU	HC tIoU	HC sIoU	Vid vIoU	Vid tIoU	Vid sIoU
Frozen Source	22.210	46.415	47.895	14.490	27.778	48.458
Projection (saved)	27.851	58.434	47.895	22.016	40.807	48.458
Original Adam + QA (saved)	31.286	60.486	47.895	22.442	40.905	48.458
Teacher Direct	22.210	46.415	47.895	14.490	27.778	48.458
Teacher IoU OPD	22.374	46.415	49.422	13.513	27.778	45.673
Teacher DIoU OPD	22.370	46.415	49.397	13.490	27.778	45.574
QA + Teacher Direct	31.286	60.486	47.895	22.442	40.905	48.458
QA + Teacher IoU OPD	31.416	60.486	49.422	21.140	40.905	45.673
QA + Teacher DIoU OPD	31.424	60.486	49.397	21.114	40.905	45.574
Offline GT Direct	28.615	46.415	61.822	23.093	27.778	79.193
Offline GT IoU OPD	27.628	46.415	59.995	23.043	27.778	77.564
Offline GT DIoU OPD	27.683	46.415	60.025	22.996	27.778	77.589
QA + Offline GT Direct	39.047	60.486	61.822	33.944	40.905	79.193
QA + Offline GT IoU OPD	37.716	60.486	59.995	33.633	40.905	77.564
QA + Offline GT DIoU OPD	37.767	60.486	60.025	33.576	40.905	77.589

All percentages are query means over the fixed four sources per direction. Original Adam + QA is reused exactly. New spatial states receive actual complete full-model re-decode; T+S combines those boxes with the locked original QA interval. Source-reference and actual-Teacher-reference results agree exactly because every selected reference is Source.

Offline GT rows are privileged mechanism diagnostics and never unlabeled method performance. Direct is an authorized comparator for this experiment, not a required paper benchmark. No best-GT-step selection or parameter search.

Reference learnability and negative tails

Case	Oracle anchors	GT Direct delta v (pp)	GT IoU delta v (pp)	GT DIoU delta v (pp)	Candidate oracle - Source IoU
HC01	3	4.129	3.892	3.906	0.000
HC02	3	9.930	8.427	8.620	0.000
HC03	1	4.700	3.143	3.013	0.000
HC04	4	6.861	6.207	6.353	0.027
V01	3	13.995	13.890	13.642	0.000
V02	4	2.405	1.700	1.680	0.000
V03	1	4.546	4.445	4.476	0.000
V04	1	13.467	14.176	14.227	0.000

Oracle uses actual GT at the unchanged four observation positions only: 20 available anchors / 12 unknown; no GT-based frame replacement. This is different from choosing the best supported candidate after GT. Original Direct planned denominator four and OPD valid-anchor mean semantics remain unchanged.

Arm	HC worst delta v	HC severe count	Vid worst delta v	Vid severe count
Teacher IoU	-0.267	0	-2.231	0
Teacher DIoU	-0.294	0	-2.227	0
GT Direct	4.129	0	2.405	0
GT IoU	3.143	0	1.700	0
GT DIoU	3.013	0	1.680	0

Severe bin is a fixed report label: delta vIoU <= -5 percentage points, not a method admission gate. All observed/unobserved dense frame effects, parameter/reference movement, positive/negative/neutral/unknown cases are included in the scalar bundle.

Compute, readback and scope

48 unlabeled fits / 240 actual optimizer steps; 24 offline-GT fits / 120 actual steps. Every fit has five saved steps, original 1,792 active Query/LN parameters and 32 antithetic Gaussian samples, matched case key and original domain configuration. DIoU changes reward only; no source-displacement cap.

Teacher input 48594 tokens / visual 36864 / generated 1248. Actual request end-to-end 35.255668s; model generation+sync 33.519341s; peak allocated GPU 17900143616bytes. Cold model load, asset hash and candidate preparation excluded. Student stage wall: method 243.860702s; offline Oracle 146.652675s; failed initial Oracle and restart custody disclosed separately.

168 official/independent full-Tube metric readbacks: maximum absolute difference 1.22e-15. All source pixels/checkpoint/frame grid/initial states bind to unchanged source evidence; actual final full-model outputs equal cached decoder boxes and frozen source weights are restored.

Oracle float32 boundary roundoff caused a saved validation error after first Direct five-step fit. Compatibility uses 1e-7 numerical encoding tolerance only, no coordinate clipping, reference or optimization change. The original Direct five-step snapshot was resumed exactly. Owned RAM high/max/swap 16/20/1GiB, disk floor 30GiB, one GPU worker at a time; no observed OOM, new tracker/teacher/download/monitor or restored historical queue.

Evidence pass means complete custody/arithmetic/output readback, not method qualification. No new teacher correction position was obtained in this input/candidate recipe; candidate restrictions and Source-first ordering remain competing explanations. Offline Oracle benefit, if any, establishes only conditional spatial learnability. Historical eight-case exposure and partial sparse GT prevent generalization or a universal route-level failure claim.